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Adversarial BEC Email Detection

AIC211 - IA2001
GROUP 3
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Leader, final integration, validation

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Literature review, methodology, MNB support

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Dataset analysis, Random Forest support

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Preprocessing, feature engineering, Logistic Regression support

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XGBoost, deployment, demo

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Robustness, Linear SVC support

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CONTENTS

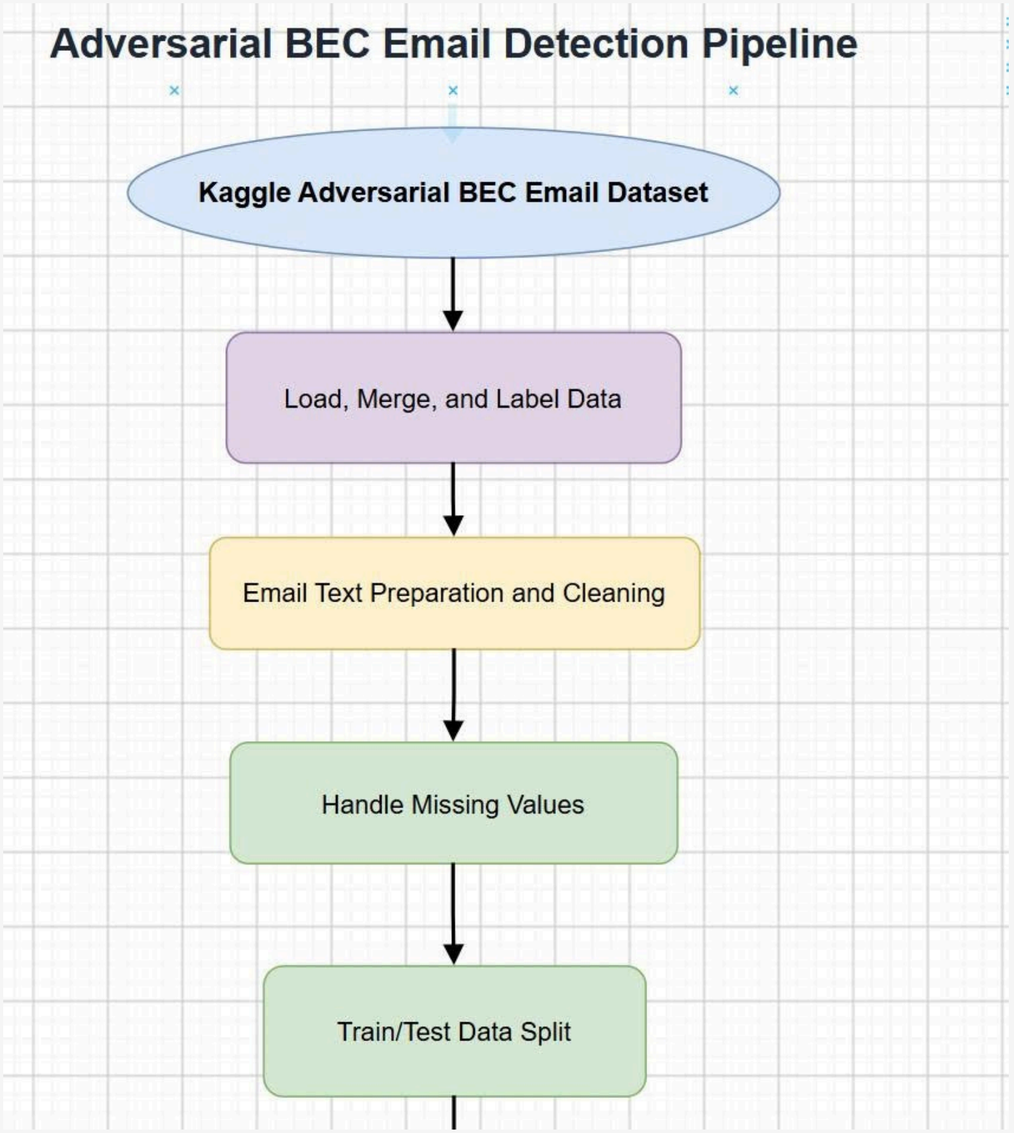
I. Pipeline

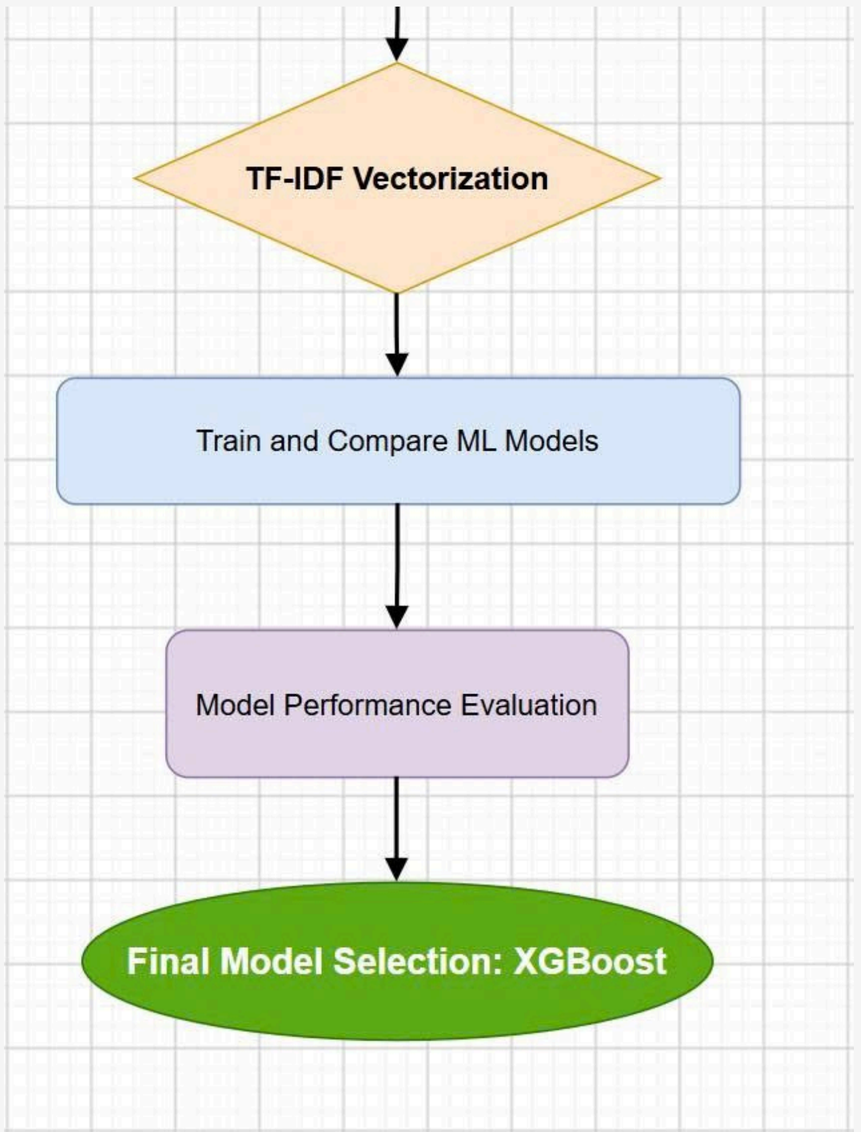
II. Dataset Overview

III. Result

IV. Conclusion & Future Work

I. PIPELINE


Adversarial BEC Email Detection Pipeline
Kaggle Adversarial BEC Email Dataset
Load, Merge, and Label Data
Email Text Preparation and Cleaning
Handle Missing Values
Train/Test Data Split


TF-IDF Vectorization
Train and Compare ML Models
Model Performance Evaluation
Final Model Selection: XGBoost

1. DATASET AND DATA PREPARATION

Kaggle Adversarial BEC Email Dataset

- Contains clean and poisoned synthetic BEC emails
- Includes email Subject and Body information
- Used to analyze adversarial changes in email text

Load, Merge, and Label Data

- Load two CSV email datasets
- Assign label 0 to clean synthetic emails
- Assign label 1 to poisoned or adversarial emails
- Merge both files into one classification dataset

2. TEXT PREPROCESSING AND DATA SPLIT EMAIL TEXT PREPARATION AND CLEANING

Email Text Preparation and Cleaning

- Handle missing values in Subject and Body
- Combine Subject and Body into one text field
- Convert text to lowercase
- Normalize punctuation and extra whitespace
- Remove stopwords and irrelevant text noise

2. TEXT PREPROCESSING AND DATA SPLIT EMAIL TEXT PREPARATION AND CLEANING

Train/Test Data Split

- Use 80% of the data for model training
- Use 20% of the data for final testing
- Apply stratified sampling based on class labels
- Preserve the class distribution in both sets

3. FEATURE EXTRACTION AND MODEL TRAINING

TF-IDF Feature Extraction

- Convert cleaned email text into numerical vectors
- Use word-level TF-IDF for traditional ML models
- Capture important words and word combinations
- Use character-level TF-IDF for XGBoost
- Capture spelling changes and adversarial patterns

3. FEATURE EXTRACTION AND MODEL TRAINING

Train and Compare ML Models

- Multinomial Naive Bayes
- Logistic Regression
- Random Forest
- Linear SVC
- Gradient Boosting
- XGBoost

4. EVALUATION AND FINAL MODEL

Model Performance Evaluation

- Accuracy – overall prediction performance
- Precision – correctness of poisoned-email predictions
- Recall – ability to detect poisoned emails
- F1-score – balance between Precision and Recall
- Confusion Matrix – analyze TP, TN, FP, and FN

Final Model Selection: XGBoost

- Achieved the best overall evaluation performance
- Provided a strong balance of Precision and Recall
- Effectively detected subtle adversarial text patterns
- Selected as the final BEC detection model

II. DATASET OVERVIEW

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Name	Kaggle Adversarial BEC Email Dataset
Nature	Synthetic dataset (Binary Classification)
Total Scale	8,422 records
Class Distribution	Balanced (2 Classes) • Class 0: Clean BEC email. • Class 1: Poisoned/Adversarial BEC email.
Attributes	Subject, Body, and Label

III. RESULT

Model	Accuracy	Precision	Recall	F1-score
Multinomial NB (Baseline)	0.8042	0.9737	0.6245	0.7609
Random Forest	0.7811	0.8276	0.7089	0.7636
Linear SVC	0.7832	0.8045	0.7468	0.7746
Logistic Regression	0.8189	0.8683	0.7511	0.8054
Gradient Boosting	0.8589	0.8571	0.8608	0.8589
XGBoost	**0.9558**	**0.9538**	**0.9578**	**0.9558**

III. RESULT

MODEL COMPARISON

6 algorithms evaluated (NB, Logistic Regression, SVC, Random Forest, Gradient Boosting, XGBoost).

TOP PERFORMER

XGBoost

- Accuracy: 0.9558
- Recall: 0.9578
- F1-Score: 0.9558

STRATEGIC CHOICE

- Model: XGBoost (primary engine).
- Priority: Maximize Recall to minimize FN, which are more critical than FP.

IV. CONCLUSION & FUTURE WORK

a. CONCLUSION

- **Pipeline:** End-to-end classification from data ingestion to automated risk scoring.
- **Core Success:** XGBoost detects text-based evasion tactics missed by signature-based systems.
- **Deployment:** Docker-based architecture ready for Proof of Concept (PoC) integration into email security gateways.

IV. CONCLUSION & FUTURE WORK

b. FUTURE WORK

- **Data Enrichment:** Integrate real-world, labeled organizational data.
- **Feature Expansion:** Add sender/recipient behavior, email headers, and temporal patterns.
- **Continuous Improvement:** Deploy automated retraining loops to mitigate drift.
- **Compliance:** Implement robust PII anonymization layers for global regulatory (GDPR) readiness.

THANK YOU